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IT4403 – Advanced Web and Mobile Applications

Milestone 1 – JSON

Due 10/5/2026

Overview

Milestone 1 demonstrates how JavaScript and jQuery can read local Open Library JSON files and display book metadata in organized webpages. I used JSONLint to validate the two JSON files provided and prettify to view the format and study the structure. The project contains one webpage for a single Open Library book-edition record and another webpage for stored results from the predetermined “modern web” search. Both pages use inline CSS and JavaScript, accessible labels, meaningful image alternative text, links to Open Library records, and fallback messages for unavailable data.

URLs

GitHub URL: [Tbar1864/IT4403Fall2026: New Website for class](https://github.com/Tbar1864/IT4403Fall2026)

Project homepage: [Projects | CCSE Information Technology](#)

Single-book webpage: [Fantastic Mr. Fox | Open Library Book Display](#)

Search-results webpage: [Modern Web Search Results | Open Library](#)

Screenshot 1: Single-book webpage

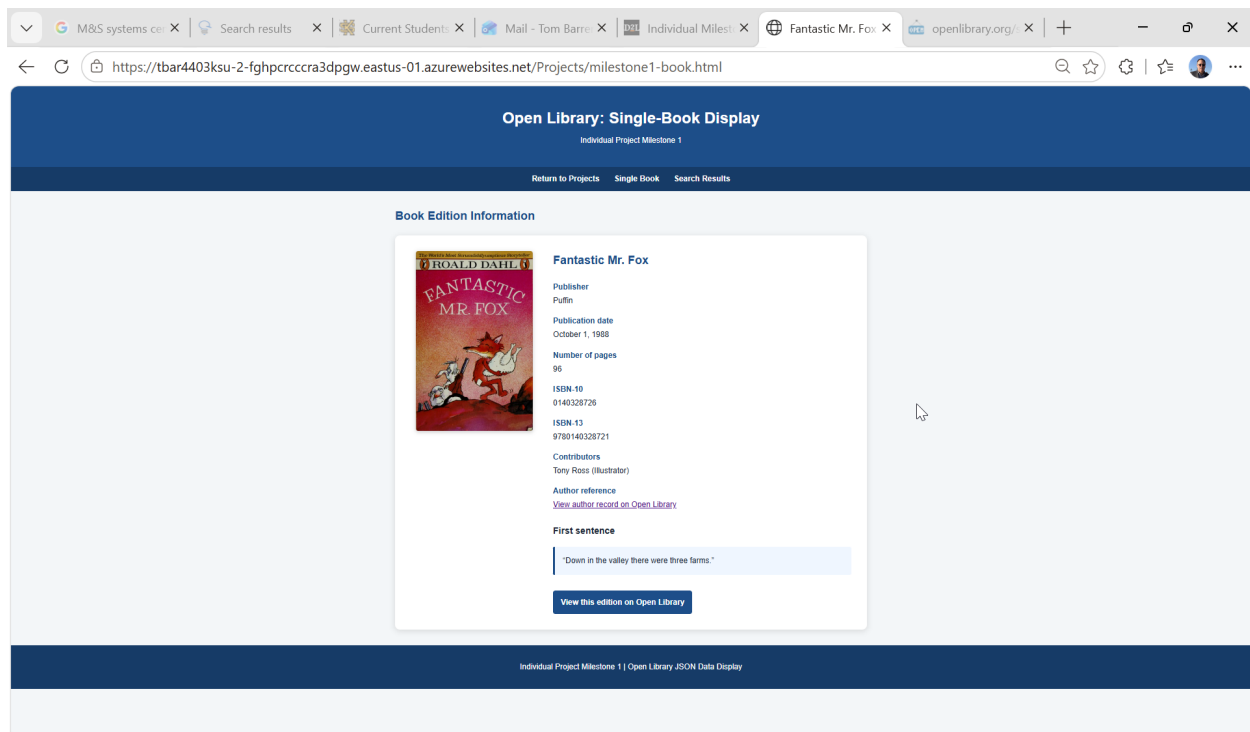


Figure 1. Single-book webpage displaying *Fantastic Mr. Fox*. This screenshot shows the webpage created from `openlibrary-book.json`. The page displays the cover image, title, publisher, publication date, page count, ISBN-10, ISBN-13, contributor, first sentence, author record link, and edition link. The JavaScript reads one book-edition object, so the page does not require a main loop. It does use array handling for properties such as publishers, covers, ISBN values, and authors.

Screenshot 2: Search-results webpage

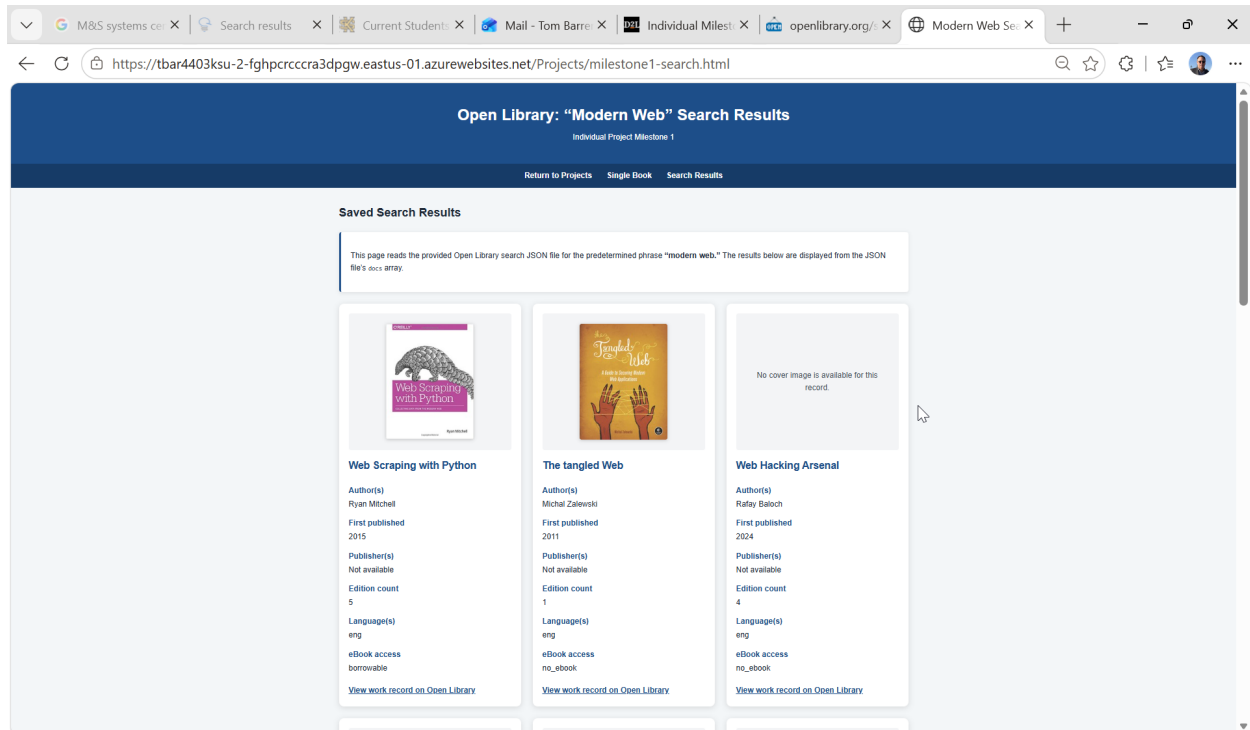


Figure 2. Search-results webpage displaying saved “modern web” results. This screenshot shows the page created from `openlibrary-search.json`. The page uses the jQuery `$.each()` method to loop through the records in the JSON file’s `docs` array. Each result is displayed in a separate card containing available title, author, first-publication year, publisher, edition count, language, eBook access, cover image, and a link to the corresponding Open Library work record.

JSON comparison

The two JSON files differ mainly in structure and purpose. The `openlibrary-book.json` file contains one Open Library edition record for *Fantastic Mr. Fox*. Its properties describe a single specific edition, including its publisher, publication date, page count, ISBNs, covers, and author reference. In contrast, `openlibrary-search.json` contains an Open Library search-response object. Its `docs` array stores multiple book/work records returned for the

search phrase “modern web.” Because the search file contains multiple records, the webpage must use a loop to process each object in the docs array.

Arrays, nested data, and missing values

The code handles arrays by using a helper function that joins available array values with commas. This is used for fields such as publishers, authors, ISBNs, contributors, and languages. The code accesses nested data by reading properties such as `first_sentence.value` and the key values contained in author and work objects. The code handles missing values safely by displaying “Not available” when a property is not present. It also displays a clear no-cover message if a book has no usable cover identifier. These safeguards prevent missing Open Library data from causing the webpages to fail.